

PIGSTI

① Preprocessing

Raw reads

(FASTQ)

② Host route

Trim adapters

(AdapterRemoval)

Host identification

(FastQ Screen)

Map host

(BWA / Bowtie2)

Map mtDNA

(BWA / Bowtie2)

Merge samples

(samtools)

Soft clipping

(Python)

Damage

(DamageProfiler)

Host species



③ Metagenomics screening

Dedup

(PRINSEQ++)

Mammals removal

(Bowtie2)

Pool samples

(cat · pigz)

Metagenome screening

(KrakenUniq)

Metagenome screening

(MALT)

④ Pathogen route

Pathogen detection

(E-value · MaltExtract)

Pathogen mapping

(BWA / Bowtie2)

Authentication

(composite score)

Authentication details

ANI

Entropy

Breadth

5' C→T damage

Edit distance

Mapping ratio

Genus rank

Verifiable

(Excel · PDF)

Genetic sexing

(sexing_residual)

- Preprocessing
- Metagenomics
- Mapping
- Optional
- Host
- Detection
- Authentication